

Code-Switching as a Controlled Variable

Hebrew-English mixing degrades sentiment classification
and displaces representations

THE PROBLEM

Israeli tech Slack is not Hebrew. It is Hebrew ⇔ English.

- Hebrew syntax, English noun — ה"deployment" הלילה עצמאב סרק
- Models are validated on clean Hebrew or clean English
- Real workplace text is neither — so reported accuracy may be optimistic
- Open question: how MUCH mixing breaks a model, and is it gradual or a cliff?

Prior work treats code-mixing as binary. We make it a dial.

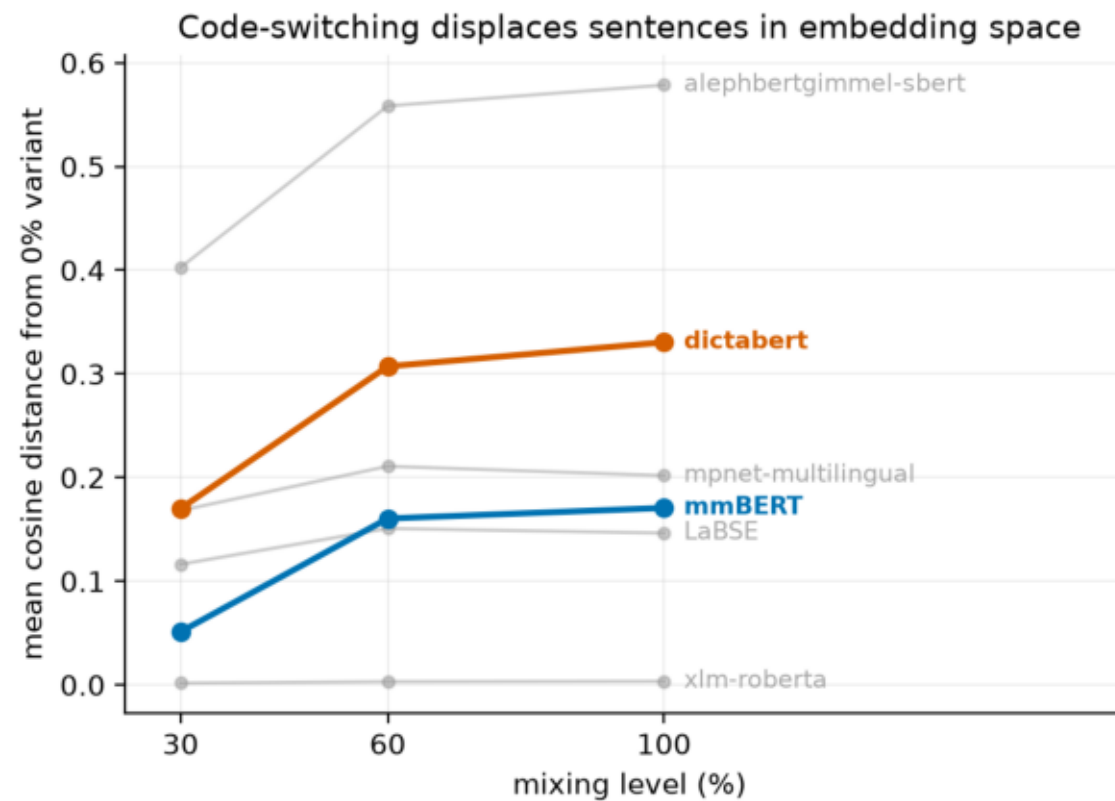
- SemEval-2020 Task 9 (SentiMix): Hinglish/Spanglish sentiment — mixed or not mixed
- GLUECoS / LinCE: standard benchmarks — no Hebrew-English
- Our framing: hold meaning FIXED, vary mixing 0 → 30 → 60 → 100%
- Turns “harder” into a dose-response curve that localises the failure

DATASET

105 seeds × 4 mixing levels = 420 items, meaning held constant

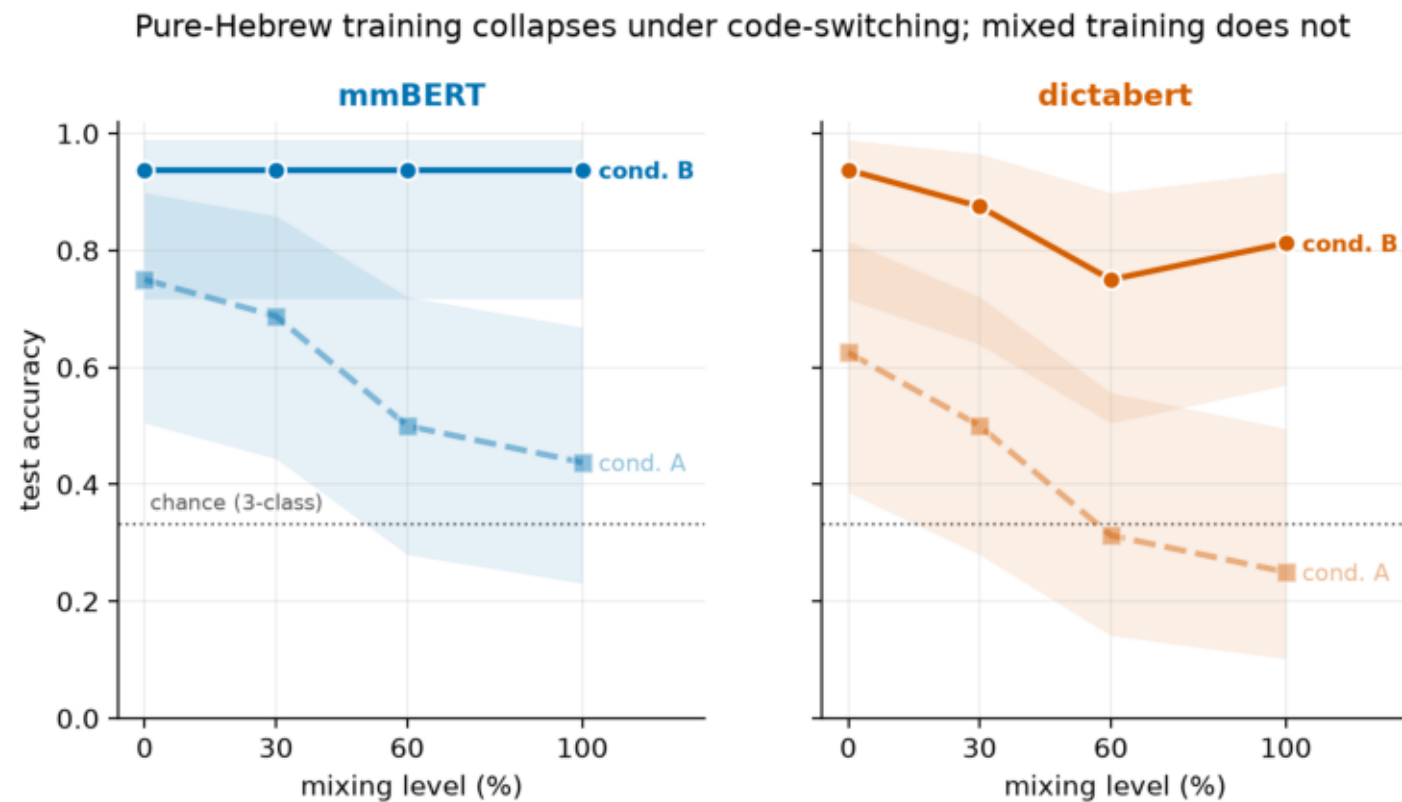
- Israeli tech registers: deploys, standups, code review, incidents, hiring
- 0% pure Hebrew · 30% English nouns · 60% frequent alternation · 100% English
- Mixing level is a manipulated variable, not a corpus property
- Sentences: LLM-generated from hand-written seeds, not scraped Slack
- Labels: human reviewer + LLM reviewer (Claude), with the human adjudicating disagreements

Mixing displaces sentences in embedding space



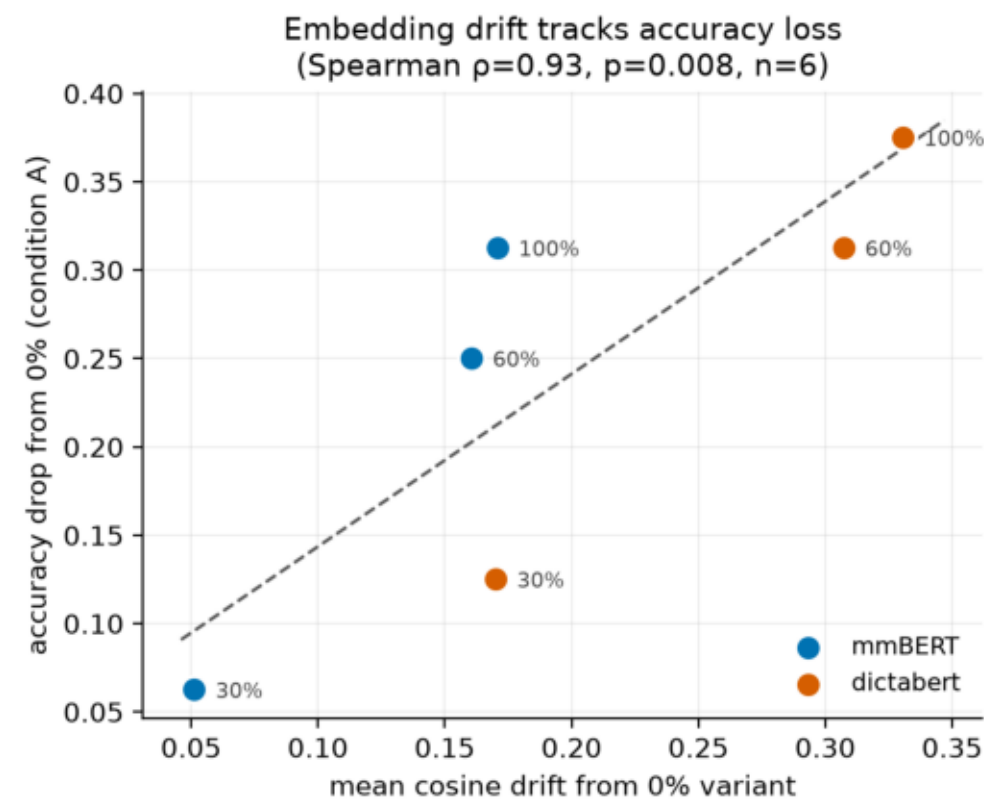
DictaBERT (Hebrew-specific) drifts ~2× further than multilingual mmBERT.

Pure-Hebrew training collapses; mixed training does not



Cond. A degrades significantly ($p=.042$ / $p=.019$). Cond. B: trend gone.

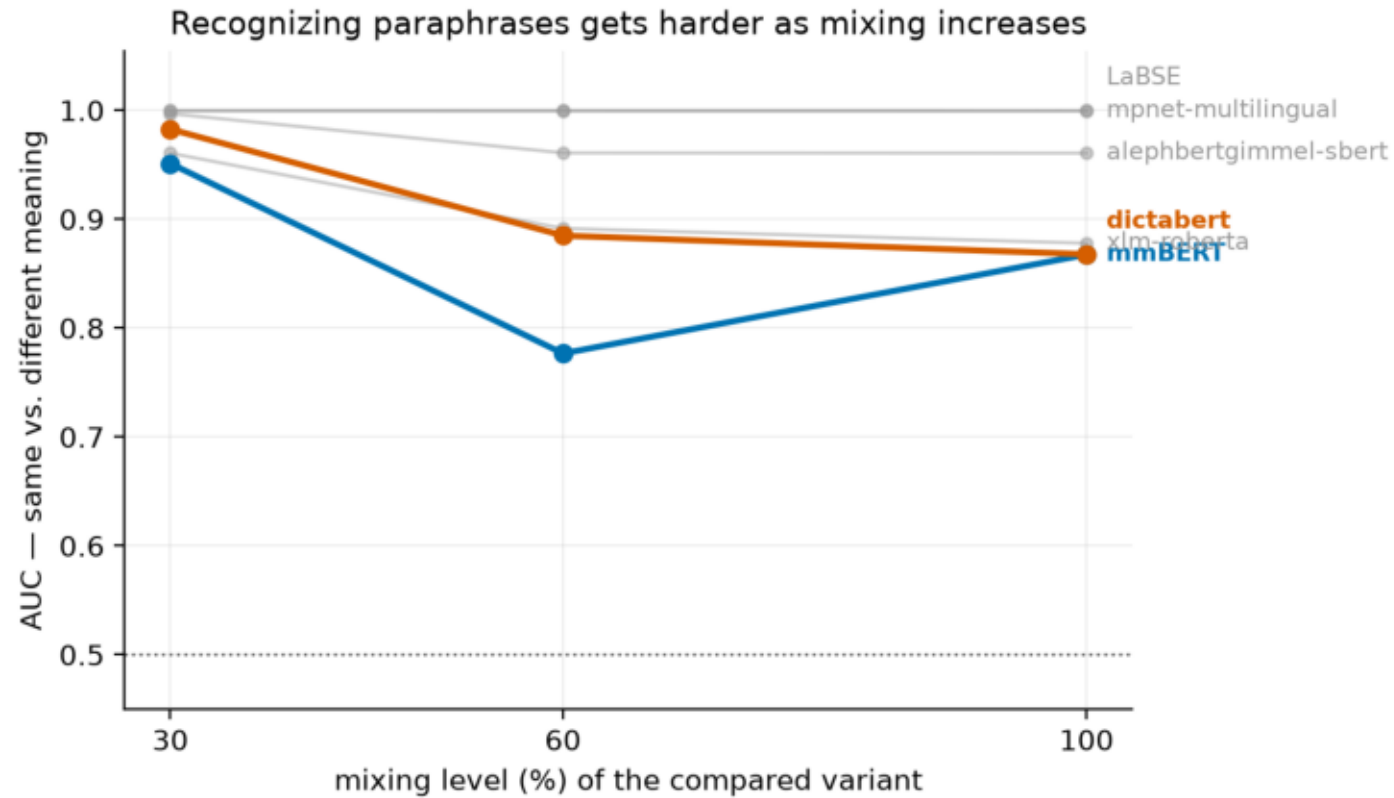
Drift predicts accuracy loss



$\rho=0.93$ ($n=6$, descriptive) — and the model that drifts more degrades more.

IS IT JUST SENTIMENT?

Recognizing paraphrases gets harder too



No new training needed. Same 30-60% break point shows up again.

Five findings

- The break is between 30% and 60% mixing — lexical borrowing is fine, structural alternation is harder
- Hebrew-SPECIFIC is WORSE than multilingual here: specialisation appears to be a liability under code-switching
- Mixed-data fine-tuning recovers it: 0.59→0.94 (mmBERT), 0.42→0.84 (DictaBERT), $p < 1e-5$
- Embedding drift is consistent with a contributing mechanism behind the accuracy loss
- The same break shows up in a complementary, label-free test (same-meaning check) — not a fluke of one task

Takeaway

A Hebrew-tuned model validated on clean Hebrew will overstate its performance on real Israeli workplace text.

A simple and effective fix in our setting is to put mixed-language data in the training set.

Code, data, and analysis notebook: see repository